EC560 Individual Project

BLP Airline Demand Data Construction

Name: Baboucarr Dibba
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GitHub repo link: https://github.com/B-Dibba/EC560-Individual-Project

AI Disclosure: Claude (Anthropic) was used to assist with R code debugging (Section 7 to 11 in the R script) and for the construction and understanding of the IV instruments used. All data construction choices (except for the IV instruments), motivations, and interpretations are my own drawn from lecture notes. Full input/output logs are included in the repository.

**Data Source**

As recommended, I use the BTS Origin and Destination Survey (DB1B), Q1 2025, Market table. DB1B is a 10% systematic sample of all domestic itineraries reported by US carriers to the DOT. The Market table records each directional origin–destination market within a multi-leg itinerary, making it the natural unit for route-level demand analysis. The Coupon and Ticket tables were loaded as well and used for inspection but are not used in construction, as the Market table contains all required prices, quantity, distance, and carrier variables.

Downloaded from the link: https://transtats.bts.gov/DatabaseInfo.asp?QO_VQ=EFI

Note: Raw DB1B CSV files (Market: around 1.2 GB, Coupon: 1.8 GB, Ticket: 0.7 GB) exceed GitHub's 100 MB file size limit and are therefore excluded from the repository via .gitignore. They can be downloaded directly from the BTS TranStats portal at the link provided above.

**Market and Product Definition**

Market: A is a directional city-pair (OriginCityMarketID and DestCityMarketID). I use city market IDs rather than airport IDs to consolidate airports serving the same metropolitan area (e.g., JFK/LGA/EWR all map to New York), consistent with the DOT's own recommendation. Markets are directional. So, SEA to LAX and LAX to SEA are separate markets, reflecting asymmetric demand and pricing.

Product: Ticketing carrier (TkCarrier) and route. I use TkCarrier, which is the brand printed on the consumer's ticket, rather than OpCarrier. The operating carrier is often an invisible regional subcontractor (e.g., SkyWest operating as "Delta Connection"), consumers choose the ticketing brand. The DB1B README confirms that TkCarrier = "99" for interline itineraries (multiple ticketing carriers), which are not attributable to a single consumer brand and are excluded.

Carrier whitelist (12 carriers): AA, AS, B6, DL, F9, G4, HA, NK, SY, UA, WN, XP. Three categories are excluded:

(1) the "99" interline placeholder.

(2) foreign carriers (AC, BA, QF, etc.) whose DB1B records represent domestic segments of international itineraries because consumers buying Paris to JFK to LAX tickets were not choosing in the US domestic market.

(3) fringe carriers (3M, MX, 9K, etc.) with negligible independent pricing power, whose demand is absorbed into the outside good. Excluded carriers represent only 1.63% of sampled passengers.

**Sample Restrictions:** Below is a table summarizing the filtration and motive.

| **Filter** | **Criterion** | **Motivation** |
|---|---|---|
| Geography | OriginCountry = DestCountry = "US" | Domestic markets only |
| Bulk fares | BulkFare = 0 | Charter/group fares are not individually priced; no consumer choice |
| Connections | MktCoupons ≤ 2 | Nonstop or one-stop; longer itineraries involve complex substitution patterns outside the model |
| Price floor | MktFare ≥ $25 | Removes data errors and award tickets reported at nominal cost |
| Price ceiling | MktFare ≤ $2,500 | Removes ultra-premium outliers that likely reflect data errors or charter mislabeling |
| Thin markets | sum(passengers) ≥ 90 per market | < 900 actual travelers/quarter produces unreliable passenger-weighted average fares (Peters 2006) |
| Monopoly markets | n_products > 1 per market | Single-carrier markets provide no within-market price variation. IV1 collapses to zero, and there is no cross-product Bertrand interaction to identify |

After all filters: 36,420 products across 8,476 markets and 12 carriers.

**Market Size and Shares**

DB1B is a 10% sample, so I expand by 10× to approximate actual passengers. Following Peters (2006) and Ciliberto & Tamer (2009), I set market size = actual_pax × 3, implying the outside good (not flying, driving, or not traveling) captures two-thirds of the potential market, which is consistent with low route-level penetration rates typical in domestic aviation. Under a constant multiplier, the outside share $s_0 = 1 - 1/3 = 0.667$ is identical across markets and is therefore absorbed by market fixed effects or the intercept in the logit regression (as discussed in Lecture 5). The Berry (1994) logit left-hand side, $\ln(s_j) - \ln(s_0)$, ranges from −11.4 to −0.69, with all values negative as required ($s_j < s_0$ everywhere).

**Instrumental Variables**

Since price is endogenous because carriers observe unobserved quality $\xi_j$ before setting fares, generating upward bias in OLS (as discussed in class). I construct three instruments:

**IV1:** BLP rival log-distance (Berry, Levinsohn & Pakes 1995). Sum of rivals' log non-stop miles on the same route. Relevance*:* fuel cost scales with distance, so rivals' distances shift rivals' marginal costs and hence prices, feeding into carrier j's Bertrand best-response. Exclusion*:* physical geography is determined by city locations, not by demand shocks for a specific carrier.

**IV2:** Hausman cost shift (Hausman, Leonard & Zona 1994; Nevo 2001): Carrier j's average fare across all other routes it serves. Relevance*:* system-wide cost shocks (jet fuel, labor contracts) shift fares uniformly across all of a carrier's routes. Exclusion*:* route-specific demand shocks are uncorrelated across routes under the no-common-demand assumption.

**IV3:** Southwest presence dummy (Goolsbee & Syverson 2008): = 1 if Southwest (WN) operates in market m and carrier j ≠ WN. Relevance: the well-documented "Southwest Effect" depresses rivals' fares by 17–26%, consistent with Bertrand best-responses to a low-cost entrant. Exclusion: Southwest's route entry is driven by gate availability and fleet constraints, not by route-specific demand shocks for legacy carriers. 56.4% of non-WN carrier-routes face Southwest competition.

**Final Dataset Summary**

| **Statistic** | **Value** |
|---|---|
| Markets (origin × dest) | 8,476 |
| Products (carrier × route) | 36,420 |
| Carriers | 12 |
| Avg. products/market | 4.3 |
| Price range | $26.87 – $2,305 |
| Share range | $7\times10^{-6}$ – 0.333 |
| Outside share | 0.667 (= 1 − 1/3) |
| y_logit range | −11.4 – −0.69 |

The largest carriers by route coverage are AA (7,997), DL (7,650), and UA (7,592). ULCCs (F9, NK) show the lowest average prices ($96, $95) and shares, consistent with their ULCC positioning.

The final dataset is saved to data/processed/blp_airline_data.csv.